

Written By Doctor Of Pharmacy

GSc101 Paper

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Question no 1:

Major Types of Ecosystem in Pakistan

Answer:

Major ecosystems in Pakistan

- Pakistan has a variety of climate and seasons.
- Have these main ecosystems:
 - o Temperate deciduous forests – Shogran and Neelum valley
 - o Coniferous alpine and boreal forests – Kaghan, Malam Jabba, Dir and Chillas
 - o Grassland ecosystems – Gilgit, Kashmir, Waziristan, Chitral and Kallat
 - o Desert ecosystem – Mianwali, Bahawal Nagaar, Bahawalpur and more, Thal, Thar and Cholistan
 - o Tundra ecosystem – mountains of Karakoram and Hindukash

Question no 2:

Formation of nitric acid By industry method?

Answer:

Preparation of Nitric acid:

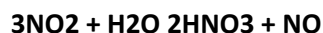
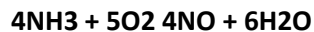
Nitric acid is prepared industrially by a method known as Ostwald's method.

In this method nitric acid is prepared by the oxidation of ammonia.

There are three main steps:

1. Oxidation of ammonia to nitric oxide
2. Oxidation of nitric oxide to NO₂.
3. Reaction of NO₂ with water to produce nitric acid.

Chemical reactions are shown by the chemical equations as follows:



Question no 3

Composition Of Blood?

Ans:

Composition of blood

Human blood consists of plasma (fluid) and cellular content (cells). Plasma is the fluid with dissolved and un-dissolved materials. It consists mainly of water, which have many inorganic and organic materials dissolved in it. Cellular content consist of Red Blood Cells (RBCs), White Blood Cells (WBCs) and Platelets. Red blood cells are also called erythrocytes and white blood cells are also called leukocytes. Plasma consists of 55% of the blood. Cells constitute 45% of the blood Average human body has about 5liters of blood.

Blood contents could be separated based upon their densities and weights by various methods including centrifugation and sedimentation

Question no 4

What is Endoplasmic reticulum ?

Endoplasmic Reticulum

Endoplasmic reticulum (ER) is a network of interconnected channels present inside the cells. This is of two types: Rough ER and Smooth ER. Rough endoplasmic reticulum is the rough due to the ribosomes present on its surface. This type is involved in protein modification. The other type is free of ribosomes so that shape of this one is smooth giving it its name. SER is involved in metabolism of lipids and carbohydrates.

Question no 5

Difference Between Cellular and Organism Respiration?

Difference between cellular and organismic respiration

- Cellular respiration is a series of events of metabolic reactions for production of energy.

- Organismic respiration is the process of coordinated movements of body that results in inhalation and exhalation of air. We also call it breathing or ventilation.

Question no 7

Define Ecosystem?

Ecosystem

An area where the organisms are living in interactions with each other and their physical surroundings is called an ecosystem. There are many types of ecosystems, mainly divided into two categories:

- o Aquatic ecosystems (water ecosystems)
- o Terrestrial ecosystems (land ecosystems)

Ecosystem

Ecosystem consist of biotic (living) and abiotic (non living) components.

Question no 8

Write about aquatic ecosystem and Terrestrial ecosystem?

Terrestrial Ecosystems

Plants and animals adapts to the changes from water to land habitat. Supporting tissue developed for the purpose; animals have skeleton and plants have vascular bundles to support these organisms on land. Water conservation is another adaptation to conserve water.

Aquatic Ecosystems

Water is a liquid medium to support life. Aquatic ecosystems may be of fresh water or marine; a lake or a sea.

Question no 9

Given the Food chain in lake ?

Answer:

Phytoplankton>Crustaceans>Fishes>Fishes

Question no 9

Write Properties Of Colloid

Answer:

This is the heterogeneous and stable mixture of particles in the solvent.

ii. Particles are larger sized and visible through the light microscope.

Particles are remaining scattered does not settle down longer period. They show Brownian motion and exhibit Tyndall effect.